Gaussian prediction and recovery: Hermite rank, observation dimension and temporal obstructions

Project Stardust

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Abstract

For a square-integrable function of one standard Gaussian coordinate, conditioning on any Gaussian observation family transports its Hermite expansion to the orthogonal projection of that coordinate. We reconstruct the full L^2 prediction formula, exact prediction-energy series and a quantitative leading-rank remainder. Applied to the actual full past of fractional Brownian motion, these identities give explicit lower bounds with exponent $2m\min(H, 1-H)$ for any nonzero Hermite coefficient of rank m, and exclude higher-power increment bounds at zero information. The inverse-number potential divides positive-rank coefficients by their rank, so it preserves this obstruction despite being compact on a single Gaussian observation space. A concrete quarter-Hurst covariance witness also separates full-past prediction from current-value prediction. An exact multi-target quartic calculation gives independent targets whose shared-observation residuals have negative cross-covariance, so block error cannot be assembled from diagonal residual energies alone. Exact finite-block retained variances include the total-variance correction terms; shifted scalar quartic variance has a strict minimum at zero shift. The projection and Hermite mechanisms are classical, and the two fractional covariance regimes are already known in primary literature. This paper supplies complete proofs and exact frozen formal correspondences; no priority claim is made. The same method controls terminal-aware Brownian bracket errors for bounded Borel weights. We also reconstruct the full joint source/sample recovery identity, compact recovery exactly at finite observed source dimension, and a positive-chaos compactness criterion. Infinite observed dimension has distance one from compact recovery approximants; even one-source compact recovery has divergent squared image energy. An accelerated Mehler limit retains full-past Markov kernels while losing initial continuity and a dense right norm-generator domain.

1 Setting and main prediction formula

Let $X : \mathcal{H} \to L^2(\Omega, P)$ be an isonormal real Gaussian family on a real separable Hilbert space: X is linear, centered, jointly Gaussian, and $\mathbb{E}[X(h)X(k)] = \langle h, k\rangle$. Fix $u \in \mathcal{H}$ with $\|u\| = 1$. Let $(h_i)_{i\in I}$ be any observation family, and put

$$M = \overline{\operatorname{span}}\{h_i : i \in I\}, \quad p = P_M u, \quad v = \|p\|^2 \in [0,1], \quad \mathcal{A} = \sigma(X(h_i) : i \in I).$$

We use L^2 equivalence classes; observation sigma-algebras are not identified literally when they differ on null sets.

Define Wick polynomials $W_n(h)$ by the generating identity

$$\exp(tX(h) - t^2\|h\|^2/2) = \sum_{n=0}^{\infty} \frac{t^n}{n!} W_n(h). \tag{1}$$

Thus $W_0(h) = 1$, $W_1(h) = X(h)$ and $W_2(h) = X(h)^2 - \|h\|^2$. At unit u, these are the probabilists' Hermite polynomials $\mathrm{He}_n(X(u))$.

Theorem 1.1. *Every $f \in L^2(\sigma(X(u)), P)$ has a unique expansion*

$$f = \sum_{n\geq 0} a_n W_n(u), \qquad a_n = \frac{\mathbb{E}[fW_n(u)]}{n!}, \qquad \|f\|_2^2 = \sum_{n\geq 0} a_n^2 n!. \tag{2}$$

Conditioning gives the norm-convergent series and exact energy

$$\mathbb{E}[f \mid \mathcal{A}] = \sum_{n\geq 0} a_n W_n(p), \qquad \|\mathbb{E}[f \mid \mathcal{A}]\|_2^2 = \sum_{n\geq 0} a_n^2 n! v^n. \tag{3}$$

If $a_n = 0$ for $n < m$, and $A_m = a_m^2 m!$, then

$$A_m v^m \leq \|\mathbb{E}[f \mid \mathcal{A}]\|_2^2 \leq v^m \|f\|_2^2, \tag{4}$$

$$0 \leq \|\mathbb{E}[f \mid \mathcal{A}]\|_2^2 - A_m v^m \leq v^{m+1}\|f\|_2^2. \tag{5}$$

For any varying observation families with $v > 0$ and $v \to 0$,

$$\frac{\|\mathbb{E}[f \mid \mathcal{A}]\|_2^2}{v^m} \longrightarrow A_m, \qquad 0 \leq \frac{\|\mathbb{E}[f \mid \mathcal{A}]\|_2^2}{v^m} - A_m \leq v\|f\|_2^2. \tag{6}$$

The exact nonlinear prediction mechanism is established Gaussian theory. McElroy and Das [3] give Hermite-based nonlinear prediction formulas for Gaussian transformations with finite data. Here the Hilbert projection describes arbitrary observation families and makes the L^2 completion and rank remainder explicit. This is a formalization exposition, not a claim to have discovered nonlinear Gaussian prediction.

2 Completeness and the conditional Wick identity

Lemma 2.1. *For all $h, k \in \mathcal{H}$ and nonnegative integers n, m,*

$$\mathbb{E}[W_n(h)W_m(k)] = \mathbf{1}_{n=m}\, n!\langle h, k\rangle^n.$$

The polynomials $W_n(u)$ have dense linear span in $L^2(\sigma(X(u)), P)$.

Proof. The Gaussian exponential formula gives

$$\mathbb{E}\left[e^{sX(h)-s^2\|h\|^2/2} e^{tX(k)-t^2\|k\|^2/2}\right] = e^{st\langle h,k\rangle}.$$

Taking coefficients yields orthogonality and its exact constant. Coefficient extraction under expectation is valid: on any compact set of real or complex parameters, a Gaussian exponential with a fixed larger real linear coefficient dominates each differentiated expression in L^1, by the Gaussian exponential moment formula.

To prove completeness rather than assume it, identify the terminal space with $L^2(\gamma)$, γ standard Gaussian measure. If g is orthogonal to all polynomials, consider $F(z) = \int g(x)e^{zx}\, d\gamma(x)$. Cauchy–Schwarz and Gaussian exponential moments make this function entire, with derivatives obtained under the integral uniformly on compact parameter sets. All its Taylor coefficients vanish, so F is identically zero. In particular the finite signed measure $g\gamma$ has zero Fourier transform. For completeness, Fourier uniqueness here follows by Gaussian convolution: Fourier inversion for a

Gaussian and Fubini show each Gaussian convolution of this measure is zero; taking an approximate identity gives zero integrals against every compactly supported continuous function, and hence the measure is zero. Thus $g = 0$. The Wick polynomials span all polynomials, because their leading coefficients are one. The orthogonal complement is therefore zero, proving density and, by Hilbert-space orthogonal expansion, (2). □

Lemma 2.2. $\mathbb{E}[W_n(u) \mid \mathcal{A}] = W_n(p)$ *for every* n.

Proof. Since $u = p + (u - p)$ and $u - p \perp M$, $X(u - p)$ is independent of every finite vector of observations: their joint Gaussian characteristic function factors because all cross-covariances are zero. A pi-lambda argument extends independence to $\mathcal{A}$. Also $X(p)$ is $\mathcal{A}$-measurable modulo null sets. Indeed p is a norm limit of finite linear combinations of observation directions, so $X(p)$ is an L^2 limit of measurable variables and has a measurable representative. Consequently

$$\mathbb{E}[e^{tX(u)-t^2/2} \mid \mathcal{A}] = e^{tX(p)-t^2\|p\|^2/2}.$$

The residual exponential has expectation one, giving the equality. The exponential series is analytic in L^2 on compact parameter sets: the same Gaussian domination as in Lemma 2.1, now using squared norms, justifies differentiating. Conditional expectation is a bounded linear map on L^2, so differentiation commutes with it. Taking the nth derivative at zero proves the claim. □

Proof of Theorem 1.1. Lemma 2.1 gives the terminal expansion and Parseval. Conditional expectation is an L^2 contraction, so it commutes with its partial-sum limit. Lemma 2.2 yields the first formula in (3). The terms are mutually orthogonal and have squared norms $a_n^2 n! v^n$; taking norms of finite sums and passing to the limit gives the energy formula. If the coefficients below m vanish, all terms start at m. Since $0 \le v \le 1$, their sum is at most $v^m \sum_n a_n^2 n!$, and at least its mth term. After subtracting that term, every remaining power is at most v^{m+1}, proving (5). For $v > 0$ division by v^m proves the last bound in (6); the squeeze theorem proves its limit. □

If $v = 0$, the predictor is $a_0 = \mathbb{E}f$, with $0^0 = 1$ in the series. No normalized-rank quotient is then used. If f is centered and nonzero, Parseval ensures a least positive m with $a_m \neq 0$; this is its Hermite rank. The upper bound in (4) is attained for a pure rank-m polynomial. No assumption that f itself is polynomial is needed anywhere.

3 Actual fractional pasts and the two covariance regimes

Let B^H be a continuous fractional Brownian motion with $0 < H < 1$, normalized by

$$R_H(s,t) = \mathbb{E}[B_s^H B_t^H] = \tfrac{1}{2}(s^{2H} + t^{2H} - |t-s|^{2H}), \qquad s,t \ge 0.$$

It has the Gaussian Hilbert realization $B_t^H = X(u_t)$, $\langle u_s, u_t \rangle = R_H(s,t)$. In particular $\|u_1\| = 1$. For $0 \le t \le 1$ put

$$\mathcal{F}_t^H = \sigma(B_s^H : 0 \le s \le t), \quad P_t = \mathbb{E}[\cdot \mid \mathcal{F}_t^H], \quad v_H(t) = \|P_t B_1^H\|_2^2, \quad \kappa_H = 2\min(H, 1-H).$$

These are conditional expectations onto the entire actual observed past. In the frozen construction a continuous modification realizes the Hilbert coordinates. Coordinatewise almost-sure equality suffices to identify their L^2 observation spaces: the sets agreeing modulo null sets form a sigma-algebra containing each coordinate preimage. Closure under countable unions only uses countable unions of null sets. Thus the augmented observation sigma-algebras agree, and the conditional expectations agree as L^2 classes. No common exceptional set for an uncountable family or literal equality of raw sigma-algebras is assumed.

Proposition 3.1. *For $0 < t \le 1$,*

$$v_H(t) \ge \tfrac{1}{4} t^{\kappa_H}. \tag{7}$$

Consequently, for $f \in L^2(\sigma(B_1^H), P)$ with terminal Hermite coefficients a_n and every nonnegative integer m,

$$\|P_t f\|_2^2 \ge \frac{a_m^2 m!}{4^m} t^{\kappa_H m}. \tag{8}$$

If f is centered, $a_m \neq 0$, and $\alpha > \kappa_H m$, then for every real C and every $\delta > 0$ some $0 < t < \min(\delta, 1)$ satisfies

$$C t^\alpha < \|P_t f - P_0 f\|_2^2. \tag{9}$$

Proof. The current observation is one member of the past. Orthogonality of conditional projection therefore gives

$$\langle P_t B_1^H, B_t^H \rangle = R_H(1,t), \qquad v_H(t) \ge R_H(1,t)^2 / t^{2H}.$$

Write $r = R_H(1,t) = (1 + t^{2H} - (1-t)^{2H})/2$. Since $(1-t)^{2H} \le 1$, $r \ge t^{2H}/2$. Also $r \ge t/2$: if $2H \le 1$ use $t^{2H} \ge t$ in the preceding bound, while if $2H \ge 1$ use $(1-t)^{2H} \le 1 - t$ in the exact formula. The two bounds give

$$v_H(t) \ge \tfrac{1}{4} \max(t^{2H}, t^{2-2H}) = \tfrac{1}{4} t^{2\min(H, 1-H)}.$$

Theorem 1.1 and the single nonnegative rank-m energy term now prove (8).

The zero-time past is generated by $B_0^H = 0$, so for centered f, $P_0 f = 0$. Put $K = a_m^2 m!/4^m > 0$ and $\beta = \kappa_H m$. Since $\alpha > \beta$, $C t^{\alpha-\beta} \to 0$ as $t \downarrow 0$. Choose t smaller than δ and one, with this expression less than K. Multiplying by t^β and using (8) gives (9). □

The lower constant $1/4$ is not claimed optimal, and this proof does not give an asymptotic equivalence for v_H. Sottinen and Viitasaari's primary prediction-law paper [4] gives explicit full-past conditional laws and no-information covariance asymptotics in the two regimes. Our elementary lower bound uses only the covariance and one current observation to obstruct any faster full-past decay. It does not claim discovery of those exponents or replace the explicit prediction kernel.

4 Inverse-number smoothing preserves the obstruction

Define on the one-coordinate terminal space the inverse-number potential

$$\mathcal{U} f = \sum_{n \ge 1} \frac{a_n}{n} W_n(u), \qquad f = \sum_{n \ge 0} a_n W_n(u). \tag{10}$$

It removes the mean. In the full Gaussian library this is the bounded inverse-number operator constructed as the divergence recovery map followed by its Hilbert adjoint; its action on each positive chaos is $1/n$. The diagonal formula (10) is its exact restriction, not a different regularizer.

Proposition 4.1. *$\mathcal{U}$ is bounded, self-adjoint, nonnegative and compact on $L^2(\sigma(X(u)), P)$. Each positive coefficient is nonzero after smoothing exactly when it was nonzero before smoothing; in particular the least positive Hermite rank is preserved. Moreover*

$$\|\mathbb{E}[\mathcal{U} f \mid \mathcal{A}]\|_2^2 = \sum_{n \ge 1} \frac{a_n^2}{n^2} n! v^n. \tag{11}$$

For fractional terminal f and any $m \geq 1$ with $a_m \neq 0$,

$$\|P_t \mathcal{U} f\|_2^2 \geq \frac{a_m^2 m!}{m^2 4^m} t^{\kappa_H m}. \tag{12}$$

Every higher-power increment bound as in (9) *therefore fails for $\mathcal{U}f$ as well. Centering of the original f is unnecessary.*

Proof. In the orthonormal basis $W_n(u)/\sqrt{n!}$, $\mathcal{U}$ is diagonal with eigenvalues $0, 1, 1/2, 1/3, \ldots$. These real nonnegative multipliers prove boundedness, self-adjointness and nonnegativity. Truncating after rank N gives a finite-rank operator with norm error $1/(N+1)$; hence $\mathcal{U}$ is compact. Its positive coefficients are a_n/n, so positive ranks are preserved. Apply Theorem 1.1 to this new terminal function to obtain (11), and Proposition 3.1 to get (12). The latter has positive constant and $P_0 \mathcal{U} f = 0$; the same elementary power comparison proves the increment obstruction. □

This compactness is on a *single* Gaussian observation space. It does not imply compactness of the full Gaussian potential on an infinite-dimensional source: on first chaos the potential acts as the identity, and an infinite orthonormal family has no norm-convergent subsequence. Nor does compact smoothing in Hermite rank imply better physical-time regularity of conditional forecasts. The obstruction retains its positive-rank power because the corresponding coefficient is divided by a fixed nonzero integer.

5 A concrete full-past versus current-value witness

The Hilbert projection in the prediction formula must use the correct observation family. An explicit example already distinguishes the full past from the present coordinate.

Proposition 5.1. *Let $H = 1/4$ and define*

$$F = \mathbb{E}[B_4^H \mid \mathcal{F}_2^H], \qquad C = \mathbb{E}[B_4^H \mid \sigma(B_2^H)].$$

Then $F \neq C$ in L^2, and $\|F\|_2^2 > \|C\|_2^2$.

Proof. The exact covariance formula gives

$$R_H(4,2) = 1, \quad R_H(4,1) = \tfrac{1}{2}(3 - \sqrt{3}), \quad R_H(2,1) = \tfrac{1}{2}\sqrt{2}, \quad R_H(2,2) = \sqrt{2}.$$

The one-observation projection is $C = B_2^H/\sqrt{2}$, so $\langle C, B_1^H \rangle = 1/2$. Since B_1^H is in the full past, $\langle F, B_1^H \rangle = R_H(4,1) = (3 - \sqrt{3})/2$, a different number. Thus $F \neq C$. The one-observation subspace is contained in the full-past L^2 subspace. Both are orthogonal projections of B_4^H; their projection identities give $F - C \perp C$. Pythagoras yields $\|F\|_2^2 = \|C\|_2^2 + \|F - C\|_2^2 > \|C\|_2^2$. □

This witness does not assert that fractional Brownian motion is a new non-Markovian example. It records exactly why replacing a full-past projection with the current-coordinate projection changes the theorem.

6 Raw quartic targets and cross-covariance of residuals

The rank-energy identity concerns one transformed coordinate. With several targets, their prediction errors need not be mutually orthogonal. We now give exact formulas and an actual Gaussian example showing why diagonal error estimates cannot be added without retaining the cross terms.

For an arbitrary direction $a \in \mathcal{H}$, not necessarily unit, write $s_a = \|a\|^2$ and $p_a = P_M a$. Define the centered raw fourth power

$$F_a = X(a)^4 - 3s_a^2 = W_4(a) + 6s_a W_2(a), \quad P_a = \mathbb{E}[F_a \mid \mathcal{A}], \quad R_a = F_a - P_a.$$

The Wick expansion follows by the explicit identities $W_2(a) = X(a)^2 - s_a$ and $W_4(a) = X(a)^4 - 6s_a X(a)^2 + 3s_a^2$. In particular a raw fourth power has both second and fourth chaos. It is not a pure rank-four target.

Proposition 6.1 (Quartic prediction and error)**.** *The actual conditional prediction is*

$$\begin{aligned} P_a &= W_4(p_a) + 6s_a W_2(p_a) \\ &= X(p_a)^4 + 6(s_a - \|p_a\|^2)X(p_a)^2 + 3\|p_a\|^4 - 6s_a\|p_a\|^2. \end{aligned}$$

Its full and retained energies are

$$\|F_a\|_2^2 = 96s_a^4, \qquad \|P_a\|_2^2 = 24\|p_a\|^8 + 72s_a^2\|p_a\|^4.$$

If $s_a > 0$ and $q = \|p_a\|^2/s_a$, then

$$\frac{\|R_a\|_2^2}{\|F_a\|_2^2} = 1 - \frac{3}{4}q^2 - \frac{1}{4}q^4. \tag{13}$$

Equivalently, for the missing linear fraction $r = 1 - q$, this is $(240r - 216r^2 + 96r^3 - 24r^4)/96$.

Proof. The conditional Wick identity from Lemma 2.2 is valid for any direction by the same orthogonal split, or by scaling. It gives the first prediction formula, and expansion of W_4, W_2 gives the second. Lemma 2.1 makes second and fourth chaoses orthogonal and gives $\|W_n(a)\|_2^2 = n! s_a^n$. Thus

$$\|F_a\|_2^2 = 24s_a^4 + 36s_a^2(2s_a^2) = 96s_a^4.$$

The same computation at p_a, retaining the coefficient $6s_a$, proves the prediction energy. Since conditional expectation is orthogonal projection, $R_a \perp P_a$ and $\|R_a\|_2^2 = \|F_a\|_2^2 - \|P_a\|_2^2$. Dividing gives (13); substituting $q = 1 - r$ and expanding proves the final polynomial. The second-chaos contribution is exactly the term that a pure fourth-chaos factor would miss. □

Source module: `GaussianQuarticConditionalVariance`.
Declarations: `prediction_wick`, `prediction_coeFn`, `prediction_norm_sq`, `variance_fraction` and `residual_energy`.

Proposition 6.2 (Residual covariance)**.** *For any $a, b \in \mathcal{H}$, put $c = \langle a, b\rangle$ and $d = \langle p_a, p_b\rangle$. Then*

$$\begin{aligned} \langle F_a, F_b\rangle &= 72 s_a s_b c^2 + 24c^4, \\ \langle P_a, P_b\rangle &= 72 s_a s_b d^2 + 24d^4, \\ \langle R_a, R_b\rangle &= 72 s_a s_b (c^2 - d^2) + 24(c^4 - d^4). \end{aligned}$$

The last expression need not be nonnegative, even when $c = 0$.

Proof. Expand the second/fourth Wick expressions and apply Lemma 2.1. Cross-chaos terms vanish; the remaining terms have coefficients $36 s_a s_b\, 2!$ and $4!$. This proves the first two identities. Each residual is orthogonal to every $\mathcal{A}$-measurable L^2 variable, including both predictions. Consequently $\langle F_a, F_b\rangle = \langle P_a, P_b\rangle + \langle R_a, R_b\rangle$, which proves the third identity without an assumption about the signs of cross-covariances. □

Source module: `GaussianQuarticResidualCovariance`.
Declarations: `quartic_covariance`, `prediction_covariance`, `residual_prediction_orthogonal` and `residual_covariance`.

Theorem 6.3 (Independent targets with negative residual covariance)**.** *Let G_0, G_1 be independent standard Gaussian random variables, observe only G_0, and put*

$$F_+ = (G_0 + G_1)^4 - 12, \qquad F_- = (G_0 - G_1)^4 - 12.$$

The targets F_+, F_- are centered and independent. Their predictions are the same variable

$$P = \mathbb{E}[F_+ \mid G_0] = \mathbb{E}[F_- \mid G_0] = G_0^4 + 6G_0^2 - 9.$$

For $R_\pm = F_\pm - P$, one has

$$\langle R_+, R_- \rangle = -312, \quad \|R_+\|_2^2 = \|R_-\|_2^2 = 1224, \quad \|R_+ + R_-\|_2^2 = 1824.$$

The unexplained fraction for the sum of the targets is $19/32$, strictly below the individual fraction $51/64$.

Proof. Use orthonormal directions e_0, e_1 with $X(e_i) = G_i$, take $a = e_0 + e_1$, $b = e_0 - e_1$, and $M = \operatorname{span}\{e_0\}$. Then $s_a = s_b = 2$, $c = 0$, and $p_a = p_b = e_0$, so $d = 1$. The two linear coordinates are jointly Gaussian with covariance zero: their joint characteristic function is $\exp(-(2t^2 + 2s^2)/2)$, a product. Thus they are independent, as are their fourth-power functions. Each fourth moment is $3 \cdot 2^2 = 12$, so centering is correct.

Proposition 6.1 gives the printed common predictor and

$$\|F_\pm\|_2^2 = 96 \cdot 2^4 = 1536, \qquad \|P\|_2^2 = 24 + 72 \cdot 2^2 = 312.$$

Hence each residual has energy $1536 - 312 = 1224$. Proposition 6.2 gives $\langle R_+, R_- \rangle = 0 - 312 = -312$. Expanding the squared norm of the residual sum yields $1224 + 1224 - 624 = 1824$. The full sum has variance $1536 + 1536 = 3072$ by independence, while its predictor $2P$ has variance $4 \cdot 312 = 1248$. Its unexplained fraction is therefore $1 - 1248/3072 = 19/32$, compared with $1 - 312/1536 = 51/64$ for each target. Since $19/32 = 38/64 < 51/64$, the inequality is strict. □

The mapped source additionally identifies the actual residual representatives

$$R_\pm = (G_0 \pm G_1)^4 - G_0^4 - 6G_0^2 - 3$$

and their integral product -312. This agrees with the projection argument above; it is not a covariance assigned to formal abstract energy variables. See `GaussianQuarticResidualCovariance.residual_plus_coeFn`, `residual_minus_coeFn`, `example_residual_integral`, `example_sum_fraction` and `sum_fraction_below_individual`.

The inference to avoid is an additive lower bound on a block's unexplained energy from its diagonal unexplained energies alone. The residuals are orthogonal to the observation space, but they are not thereby orthogonal to each other. Nor is their conditional covariance asserted negative at every observation value: the theorem concerns their actual integrated L^2 covariance. This is a concrete classical projection diagnostic, with no new law of Gaussian prediction or multi-target error bound claimed.

Exact finite-block retained variance

The cross terms can be retained exactly for any finite block of target directions $(u_i)_{i\in J}$. Define covariance matrices and diagonal data

$$C_{ij} = \langle u_i, u_j\rangle, \quad B_{ij} = \langle P_M u_i, P_M u_j\rangle, \quad c_i = C_{ii}, \quad b_i = B_{ii}, \quad r_i = c_i - b_i \geq 0.$$

Orthogonal projection gives $C = B + A$, where $A_{ij} = \langle u_i - P_M u_i, u_j - P_M u_j\rangle$ and $A_{ii} = r_i$. Thus A, B, C are positive semidefinite covariance matrices, but their individual off-diagonal entries need not be positive. For a real matrix D and real diagonal lists d, e, use

$$T_1(D) = \sum_{i,j} D_{ii} D_{jj} D_{ij}^2, \qquad T_2(D) = \sum_{i,j} D_{ij}^4, \qquad \mathcal{W}(d, e; D) = \sum_{i,j} d_i e_j D_{ij}^2.$$

Consider the actual Gaussian block, with its normalization fixed,

$$\mathcal{V} = \frac{1}{24}\sum_{i\in J} F_{u_i}, \qquad \widehat{\mathcal{V}} = \mathbb{E}[\mathcal{V} \mid \mathcal{A}].$$

Both are centered square-integrable variables, since the block is finite and each Gaussian target has finite moments of every order. Proposition 6.2, summed over all ordered pairs, gives

$$\begin{aligned} \|\mathcal{V}\|_2^2 &= \frac{72T_1(C) + 24T_2(C)}{576}, \\ \|\widehat{\mathcal{V}}\|_2^2 &= \frac{72\mathcal{W}(c, c; B) + 24T_2(B)}{576}. \end{aligned} \tag{14}$$

The denominator is 24^2; no independent-site approximation is used. The retained second-chaos coefficient still contains the *total* variance list c, not just the retained list b.

Expanding $c = b + r$ and using symmetry of B yields the full three-term formula

$$\begin{aligned} \|\widehat{\mathcal{V}}\|_2^2 = \frac{72T_1(B) + 24T_2(B)}{576} \\ + \frac{1}{8}\mathcal{W}(r, r; B) + \frac{1}{4}\mathcal{W}(b, r; B). \end{aligned} \tag{15}$$

Indeed bilinearity gives $\mathcal{W}(b + r, b + r; B) = \mathcal{W}(b, b; B) + \mathcal{W}(r, r; B) + 2\mathcal{W}(b, r; B)$; $\mathcal{W}(b, b; B) = T_1(B)$, and $72/576 = 1/8$. All three terms in (15) are nonnegative, because $b_i, r_i \geq 0$ and the matrix entries are squared or raised to the fourth power. The first term alone is the variance of $24^{-1}\sum_i \left(X(P_M u_i)^4 - 3b_i^2\right)$. It generally undercounts the actual retained block variance: the other two terms are the precise missing corrections.

Equivalently, writing $Y_i = X(P_M u_i)$, the actual prediction is

$$\widehat{\mathcal{V}} = \frac{1}{24}\sum_i \{Y_i^4 - 3b_i^2 + 6r_i(Y_i^2 - b_i)\}.$$

This follows by the already proved quartic conditional formula. A term involving interior variance can survive conditioning through a retained quadratic observable. Orthogonality of a single residual to the retained space does not remove residual cross covariances between different sites. In particular the earlier explicit two-target example still prevents replacing a block residual norm by its diagonal sum. All these statements concern an actual Gaussian reference. They do not supply an interacting Gibbs entropy inequality, entropy tensorization, or a rate uniform in block size.

Source correspondence: `GaussianQuarticBlockVariance`; `block_norm_sq`, `block_prediction_condExp`, `block_prediction_norm_sq`, `block_prediction_split`.

Shifted scalar fourth powers: a genuine variance minimum

For a deterministic shift $b \in \mathbb{R}$ and a centered Gaussian Z of variance $r \geq 0$, set

$$U_b = (b + Z)^4 - (b^4 + 6b^2 r + 3r^2).$$

The subtracted expression is its actual mean. If $r = 0$, $Z = 0$ almost surely and $U_b = 0$ for every shift. For $r > 0$ use the monic variance-r Hermite polynomials $H_j(Z; r)$, characterized by $\exp(tZ - rt^2/2) = \sum_j H_j(Z; r)t^j/j!$. Direct polynomial expansion gives

$$U_b = H_4 + 4bH_3 + 6(b^2 + r)H_2 + 4b(b^2 + 3r)H_1.$$

Here $H_1 = Z$, $H_2 = Z^2 - r$, $H_3 = Z^3 - 3rZ$ and $H_4 = Z^4 - 6rZ^2 + 3r^2$; substitution verifies both centering and the identity. Their orthogonality follows from the same Gaussian generating function calculation as Lemma 2.1, with squared norms $j!r^j$. Therefore

$$\begin{aligned} \mathbb{E}U_b^2 &= 24r^4 + 16b^2(6r^3) + 36(b^2 + r)^2(2r^2) + 16b^2(b^2 + 3r)^2 r \\ &= 16b^6 r + 168b^4 r^2 + 384b^2 r^3 + 96r^4. \end{aligned} \tag{16}$$

Every shift-dependent term is nonnegative, and for $r > 0$, $b \neq 0$, the term $384b^2r^3$ is strictly positive. Thus the variance has its unique minimum $96r^4$ at zero shift when the residual variance is positive. This is a valid positive local statement, rather than an assumption about how all conditional errors combine.

For the actual Gaussian orthogonal split $X(u) = X(P_M u) + Z$, the residual $Z = X(u - P_M u)$ is independent of the entire observed sigma-algebra and has variance $r = \|u - P_M u\|^2$. Consequently the conditional variance of $X(u)^4$ is the polynomial (16) with $b = X(P_M u)$. This identifies an actual conditional variance: integrate only the independent Gaussian residual, first for bounded product tests of the observed variables and then by Gaussian polynomial integrability. For one observed coordinate, it is also the literal scalar Gaussian fiber formula. A minimum at a deterministic shift proves neither a positive lower bound for every block residual nor decay of boundary information under resampling.

Source correspondence: `GaussianShiftedQuarticVariance`; `shiftedCentered_coeFn`, `shiftedCentered_norm_sq`, `variance_minimum`, `variance_strict`, `fiber_raw_mean`, `fiber_variance`. The full-observation conditional interpretation is a labelled human application of the existing independent orthogonal split; the mapped scalar fiber statements retain their literal one-coordinate law.

7 Past-plus-terminal observations and bracket errors

The same Gaussian projection mechanism controls a distinct question with anticipative coefficients. Let W be standard Brownian motion under its original law and fix $T > 0$. Write

$$\mathcal{G}_s = \sigma(W_r : 0 \leq r \leq s) \vee \sigma(W_T), \qquad 0 \leq s < T.$$

This includes the entire raw past together with a single terminal value. It is not a claim of predictability for the original Brownian filtration. For $s \leq t \leq T$, put $h = t - s$, $L = T - s$ and $Q_{s,t} = (W_t - W_s)^2 - h$.

Proposition 7.1. *The actual conditional projection is*

$$\mathbb{E}[Q_{s,t} \mid \mathcal{G}_s] = \frac{h^2}{L^2}\{(W_T - W_s)^2 - L\}. \tag{17}$$

Consequently, for $A \in L^2(\mathcal{G}_s)$,

$$|\mathbb{E}[AQ_{s,t}]| \le \frac{2h^2}{L}\|A\|_2. \tag{18}$$

Proof. Set $S = W_T - W_s$ and $\xi = W_t - W_s - (h/L)S$. These are jointly Gaussian, with $\mathbb{E}S^2 = L$, $\mathbb{E}\xi^2 = h - h^2/L$ and $\mathbb{E}\xi S = 0$. Each increment after s is independent of the whole past up to s. Thus ξ is independent of the joint family consisting of S and that whole past: the finite joint Gaussian characteristic functions factor, and a monotone-class argument extends independence from cylinders to the generated sigma algebra. This algebra is precisely $\mathcal{G}_s$, since W_s is already observed. Expanding $((h/L)S + \xi)^2 - h$ and integrating the independent centered ξ gives (17). When $h = 0$ the same identity is zero.

The centered Gaussian square has second moment $2L^2$, since $\mathbb{E}S^4 = 3L^2$. Hence the projected variable has norm $\sqrt{2}h^2/L \le 2h^2/L$. The product with A is integrable by Cauchy–Schwarz. Orthogonal projection transfers its pairing to the conditional variable, and another Cauchy–Schwarz inequality gives (18). This is the actual finite/infinite-observation Wick projection of the mapped source, not only conditioning on the current endpoint in place of the entire past. □

Use a uniform grid $t_i = iT/n$, $0 \le i \le n$, and write $h = T/n$ and $Q_i = Q_{t_i,t_{i+1}}$. The source indexes $n = N + 1$; we use $n \ge 1$ throughout. Let $H_n = \sum_{k=1}^{n} 1/k$.

Theorem 7.2. *For any triangular array $A_{n,i} \in L^2(\mathcal{G}_{t_i})$ with $\|A_{n,i}\|_2 \le M$,*

$$\left|\mathbb{E}\sum_{i=0}^{n-1} A_{n,i}Q_i\right| \le \frac{2MT}{n}H_n \le \frac{4MT}{\sqrt{n}} \longrightarrow 0. \tag{19}$$

The harmonic-sum bound and its consequence $2MT(1 + \log n)/n$ are direct human refinements of the formally recorded last bound. No uniform boundedness or differentiability of the coefficients is required in this mean statement.

Proof. Apply (18) cell by cell. Since $T - t_i = h(n - i)$, each pairing is at most $2Mh/(n - i)$. Summing proves the first inequality. Also $1/k \le 1/\sqrt{k}$ and, by integration of $x^{-1/2}$, $\sum_{k=1}^{n} 1/\sqrt{k} \le 1 + \int_1^n x^{-1/2}dx = 2\sqrt{n} - 1 \le 2\sqrt{n}$. This gives the formally recorded estimate. Integration of $1/x$ gives $H_n \le 1 + \log n$, proving the stated refinement and convergence. Each expectation is legitimate by the pairing integrability already proved, and the sum is finite. □

Uniform bounds on the separate coefficient L^2 norms do not replace the observation-domain hypothesis. On $T = 1$, set $A_{n,i} = nQ_i$. Then the exact Gaussian square variance gives

$$\|A_{n,i}\|_2^2 = n^2(2/n^2) = 2, \qquad \mathbb{E}\sum_{i=0}^{n-1} A_{n,i}Q_i = n^2(2/n^2) = 2$$

for every n. Thus the means do not tend to zero. This coefficient uses the unresolved current increment. In particular the whole array cannot satisfy the generated-past-plus-terminal hypothesis, since Theorem 7.2 would then force its constant mean two to vanish. The source states this existential failure of the observation domain; we do not strengthen it to failure for every individual cell.

8 Strong bracket convergence for bounded Borel weights

The previous result concerns a mean, not the norm of a random sum. A separate cross-cell calculation supplies a strong statement. Let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be Borel measurable with $|\varphi| \le 1$, and define

$$a_i = \varphi(W_{t_i}, W_T), \quad B_i = a_i Q_i, \quad E_n = \sum_{i=0}^{n-1} B_i.$$

No continuity of φ is imposed.

Theorem 8.1. *The actual finite random variables satisfy*

$$\|E_n\|_2^2 \le \frac{4T^2}{n} + \frac{8T^2 H_n}{n}, \tag{20}$$

$$\|E_n\|_2^2 \le \frac{4T^2}{n} + \frac{16T^2}{\sqrt{n}} \longrightarrow 0. \tag{21}$$

The second estimate is the mapped formal theorem. The sharper harmonic estimate is a human deduction retaining the same row calculation. Thus the displayed error is $O(\sqrt{\log(n+1)/n})$ in L^2, without asserting optimality, a central limit theorem or a limiting weighted occupation integral.

Proof. Each Q_i has norm $\sqrt{2}h \le 2h$. Multiplication by a bounded measurable variable of absolute value at most one contracts L^2, so $\|B_i\|_2 \le 2h$. The diagonal sum is therefore at most $4nh^2 = 4T^2/n$.

For $i < j$, put $C_{ij} = Q_i a_i a_j$. Both endpoints of the ith increment are at most t_j. The variables a_i, a_j are measurable with respect to $\mathcal{G}_{t_j}$, and so is Q_i. Thus $C_{ij} \in L^2(\mathcal{G}_{t_j})$ and $\|C_{ij}\|_2 \le 2h$. The exact cross pairing is

$$\mathbb{E}[B_i B_j] = \mathbb{E}[Q_j C_{ij}].$$

This product is integrable: $Q_j, C_{ij} \in L^2$. Equivalently the original $B_i B_j$ is integrable by the same inequality. No independence of the weighted cells is assumed.

Define the later-time row $C_j = \sum_{i<j} C_{ij}$. It belongs to $L^2(\mathcal{G}_{t_j})$ and satisfies $\|C_j\|_2 \le 2jh \le 2T$. Expanding the exact finite sum energy yields

$$\|E_n\|_2^2 = \sum_i \|B_i\|_2^2 + 2\sum_j \mathbb{E}[Q_j C_j].$$

Apply the first bound in (19) to the row array with $M = 2T$. The absolute value of its sum of pairings is at most $4T^2 H_n/n$. Combining with the diagonal proves (20). Using $H_n \le 2\sqrt{n}$ gives (21). Its right side tends to zero; taking square roots gives the strong convergence. Finally $H_n \le 1 + \log n$ supplies the printed rate refinement. All bounded products are taken in the actual Gaussian probability space, rather than in an invented coefficient pairing. □

For any finite bound $|\varphi| \le K$, multiplication and rescaling by $K > 0$ multiply these squared bounds by K^2; $K = 0$ is immediate. The formal source uses one fixed function φ and the normalized bound one. This rescaling and the harmonic refinement are explicit classical deductions. The theorem addresses the centered bracket error alone; it does not assert that $\sum_i h\varphi(W_{t_i}, W_T)$ converges to a time integral for an arbitrary Borel function.

Anticipative variation theory has extensive classical antecedents. Yoshida's primary introduction [5] treats variation sums with random weights, explicitly allows nonpredictable weights, and sets up a substantially stronger fluctuation/expansion problem with Malliavin–Sobolev regularity. We use this as context, not as a theorem covering every bounded Borel weight here. The complete elementary Gaussian projection and later-row proof above has a narrower target, with explicit hypotheses and finite errors. No new anticipative stochastic integral or first variation convergence principle is claimed.

9 Recovery requires both sample and source projection

We now distinguish the Hermite rank of a target from the dimension of the observation space M. Work in the full Gaussian space $\mathcal{G} = L^2(\sigma(X(h) : h \in \mathcal{H}), P)$, with no additional random factor, and in its vector counterpart $\mathcal{V} = L^2(P; \mathcal{H})$ over the same sigma-algebra. Let $Q = \mathbb{E}[\cdot \mid \mathcal{A}]$ on scalars, let $\widehat{Q}$ be the Bochner conditional expectation on vectors, and let $\widehat{P}$ act pointwise by the source projection P_M. On a tensor af, with $a \in \mathcal{H}$, $f \in \mathcal{G}$,

$$\widehat{Q}(af) = aQf, \qquad \widehat{P}(af) = (P_M a)f.$$

Finite sums of such tensors are dense in $\mathcal{V}$. The two operators are commuting orthogonal projections, first on these tensors and then by continuity. Their product $J = \widehat{P}\widehat{Q}$ is therefore the orthogonal projection onto the fields that are both $\mathcal{A}$-measurable and M-valued.

Let D denote the original closed Gaussian gradient, obtained by closing the cylindrical polynomial gradient, and let $\delta = D^*$ be its maximal Hilbert adjoint. Write P_q for orthogonal projection to chaos q. With the positive number-operator convention $N = \delta D$, define N^{-1} to be zero on constants and multiplication by $1/q$ on chaos $q \geq 1$. Its associated recovery is

$$Rf = \sum_{q \geq 1} \frac{1}{q} D P_q f, \qquad \|Rf\|_{\mathcal{V}}^2 = \sum_{q \geq 1} \frac{\|P_q f\|_2^2}{q} \leq \|f\|_2^2. \tag{22}$$

This bounded map is defined for every Gaussian L^2 target; it does not require $f \in D(D)$.

We give the graph justification rather than treating the notation DN^{-1} as a domain assertion. Finite tensor Hermite polynomials obey $\delta D f_q = q f_q$ and $\langle Df_q, Dg_r \rangle = \mathbf{1}_{q=r} q \langle f_q, g_q \rangle$. These follow by one-coordinate Hermite differentiation, its Gaussian integration-by-parts adjoint, and orthogonality of the product basis. Closing these finite identities first in a fixed chaos gives $P_q f \in D(D)$, the same gradient norm and the original adjoint graph pair $(DP_q f, qP_q f)$. Equation (22) follows by orthogonal summation of the gradient fields. The finite recovery sums and their divergences converge respectively to Rf and $f - \mathbb{E}f$; closedness of δ proves

$$Rf \in D(\delta), \qquad \delta Rf = f - \mathbb{E}f. \tag{23}$$

Every such recovery field is orthogonal to $\ker \delta$: for a divergence-free v, adjointness gives $\langle v, DP_q f \rangle = \langle \delta v, P_q f \rangle = 0$. Pass through the convergent series. Any other field with the same divergence differs from Rf by a kernel element, so Pythagoras proves that Rf is the unique minimum-norm preimage.
Source correspondence: `GaussianDivergenceRecovery.recovery_graph`, `recovery_inverse_energy_hasSum`, `recovery_orthogonal_kernel`.

Theorem 9.1 (Joint observation/recovery identity)**.** *For every $f \in \mathcal{G}$,*

$$JRf = RQf.$$

In particular RQf is both observed and M-valued, and is the minimum-norm original divergence preimage of $Qf - \mathbb{E}f$. Projection of the sample variable alone generally fails: if $p = P_M u$, then for $q \geq 1$,

$$\widehat{Q} R W_q(u) - RQW_q(u) = (u - p)W_{q-1}(p), \tag{24}$$

$$\|\widehat{Q} R W_q(u) - RQW_q(u)\|_{\mathcal{V}}^2 = (q-1)!\|u - p\|^2 \|p\|^{2(q-1)}. \tag{25}$$

At $q = 1$ the defect is the deterministic field $u - p$.

Proof. The Wick derivative is $DW_q(u) = q\,uW_{q-1}(u)$, so $RW_q(u) = uW_{q-1}(u)$. The conditional identity already proved in Section 2 gives $QW_q(u) = W_q(p)$. Consequently

$$JRW_q(u) = pW_{q-1}(p) = RQW_q(u), \qquad \widehat{Q}RW_q(u) = uW_{q-1}(p).$$

Subtracting proves (24); its norm is the product of the deterministic squared norm and the Wick energy $(q-1)!\|p\|^{2(q-1)}$, proving (25). The constant case has zero recovery on both sides.

Pure Wick powers span a dense subspace of $\mathcal{G}$. To see the needed spanning statement, expand $W_q(\sum_j t_j u_j)$ as a polynomial in the parameters: its coefficients give every homogeneous Wick product of q Gaussian directions. Polarization expresses each coefficient as a finite linear combination of values of these pure powers. Tensor Hermite polynomials are products of Wick powers in orthogonal directions and have exactly these homogeneous Wick products. Finite-dimensional product Hermite bases are complete by the one-coordinate completeness proof in Section 2 and density of finite sums of product functions. Finally choose a countable dense orthonormal source family. Gaussian L^2 limits of its linear coordinates generate all $X(h)$ modulo null sets, and finite-coordinate cylindrical functions are dense in their generated L^2 space by the monotone-class approximation of indicator functions. This proves the asserted density.

Both JR and RQ are bounded, so their equality on pure Wick powers extends to all $\mathcal{G}$. The range property follows from the definition of J; (23) applied to Qf and $\mathbb{E}Qf = \mathbb{E}f$ gives its divergence. The minimum-norm conclusion was proved above. □

Source correspondence: `GaussianWickRecoveryPrediction.prediction_recovery_defect`, `prediction_recovery_defect_energy`; `GaussianJointSourceProjection.jointConditional_fixed_iff`; `GaussianJointRecoveryIntertwining.joint_recovery_operator`.

These are source/sample constraints at a fixed observation family. They do not by themselves identify an adapted physical-time stochastic integrand or a Clark–Ocone representation for a different observed process.

10 A stationary factor need not decrease defect energy

The preceding source/sample identities use specific compatible Gaussian operators. General stationary observation factors require their own energy argument. Here is the classical positive mechanism. On a real Hilbert space let T be a bounded symmetric contraction and let Q be the orthogonal projection onto a closed T-invariant subspace V. Symmetry makes $V^\perp$ invariant as well: if $r \perp V$ and $v \in V$, $\langle Tr, v\rangle = \langle r, Tv\rangle = 0$. Thus $TQ = QT$. Writing $f = Qf + r$ gives the exact splitting

$$\langle (I-T)f, f\rangle = \langle (I-T)Qf, Qf\rangle + \langle (I-T)r, r\rangle. \tag{26}$$

Since $\langle Tr, r\rangle \le \|Tr\|\|r\| \le \|r\|^2$, the residual term is nonnegative. Hence Q decreases defect energy. The isometric-factor version in the frozen library uses $Q = JJ^*$, $TJ = JS$ and the same orthogonal decomposition; it is not a separate probabilistic assumption. In particular, for each $t > 0$ the conclusion applies to $t^{-1}(I - T_t)$ when T_t is a symmetric contraction preserving the factor. Uniform upper bounds on these defect energies then transfer to the factor, as do existing finite limiting energies. No unbounded-generator domain is inferred just from one bounded defect.

Stationarity and factor invariance alone do not suffice. Take the uniform probability law on $\{0,1,2\}$ and the Markov matrix

$$T = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 \end{pmatrix}, \qquad Qf = \left(f_0, \frac{f_1+f_2}{2}, \frac{f_1+f_2}{2}\right).$$

Every row and column of T sums to one, all entries are nonnegative, and Q is exactly conditional expectation on the partition $\{\{0\},\{1,2\}\}$. Jensen's inequality and column sums prove $\|Tf\|_{L^2}^2 \le \|f\|_{L^2}^2$. The range of Q is T-invariant, since

$$T(a,b,b) = (b,(a+b)/2,(a+b)/2).$$

Its genuine coarse Markov matrix is therefore $\left(\begin{smallmatrix} 0 & 1 \\ 1/2 & 1/2 \end{smallmatrix}\right)$, with stationary masses $1/3, 2/3$. Nevertheless the original T is not symmetric: $T_{01} = 1 \neq 1/2 = T_{10}$. For

$$f = (2,-1/4,-7/4), \qquad Qf = (2,-1,-1), \qquad \mathcal{E}(f) = \frac{1}{3}\sum_{j=0}^{2}(f_j - (Tf)_j)f_j,$$

direct evaluation gives

$$Tf = (-1/4,1/8,1/8), \quad TQf = (-1,1/2,1/2), \quad \mathcal{E}(f) = 21/8 < 3 = \mathcal{E}(Qf).$$

Also $QTf = Tf \neq TQf$. Thus the symmetric splitting cannot be imported merely because the retained subspace is invariant. Both vectors are centered in the actual stationary law. Defect energy is nonnegative here, by the same contraction bound, even though projection can increase it. The conventional reversible Dirichlet-form setting is recalled in Levin–Peres [8], Section 13.2. The example tests the stated generalization; no new Markov-energy principle or novelty for its finite mechanism is asserted.

Source correspondence: `SymmetricFactorEnergy.quadratic_energy_split`, `SymmetricDefectEnergy.defect_energy_contraction`; `NonreversibleFactorCounterexample.retained_subspace_invariant`, `energy_witness`, `energy_projected_witness`.

11 Finite observed dimension and compact recovery

Theorem 11.1 (Two different compactness classifications)**.** *For an arbitrary observation index set I,*

$$Q \text{ is compact} \iff M = \{0\}, \qquad RQ \text{ is compact} \iff \dim M < \infty.$$

If $m = \dim M < \infty$ and $S_N = \sum_{q=0}^{N} P_q$, then RS_NQ has finite rank at most $(N+1)^m$, and

$$\|RQ - RS_NQ\| \le (N+1)^{-1/2}. \tag{27}$$

The combined rank bound is an elementary consequence of the finite Hermite construction below; the source gives the corresponding per-chaos bounded-Hermite-space dimension and the recovery tail estimate.

Proof. If $M \neq 0$, choose a unit $u \in M$. The normalized Wick variables $W_q(u)/\sqrt{q!}$, $q \ge 0$, are orthonormal and fixed by Q. They have no norm-convergent subsequence, so Q is not compact. If $M = 0$, every observation is zero, $Qf = \mathbb{E}f$ is a rank-one map, and $RQ = 0$.

Suppose $\dim M = m < \infty$, and choose an orthonormal basis $u_1,\dots,u_m$. Their Gaussian coordinates are independent standard normals. The observation L^2 space has product Hermite basis

$$H_\alpha = \prod_{j=1}^{m} \frac{\mathrm{He}_{\alpha_j}(X(u_j))}{\sqrt{\alpha_j!}}, \qquad \alpha \in \mathbb{N}^m.$$

This is the same observed space even if infinitely many labels or a singular redundant covariance matrix describe it: source-norm convergence and Gaussian L^2 limits identify the generated sigma-algebras modulo null sets. The product H_α belongs to chaos $|\alpha|$. Thus $S_N Q$ takes values in the span of the finitely many H_α with $|\alpha| \le N$. Each $\alpha_j \le N$, so there are at most $(N+1)^m$ such indices. Applying R cannot increase rank.

The chaos decomposition is preserved by Q, as follows on pure Wick powers from their conditional identity and then by density. Using (22),

$$\|R(I - S_N)Qf\|^2 = \sum_{q>N} \frac{\|P_q Q f\|^2}{q} \le \frac{1}{N+1}\|Qf\|^2 \le \frac{1}{N+1}\|f\|^2.$$

This proves (27). A norm limit of finite-rank operators is compact: their images give finite nets for the limiting image of the unit ball, which is relatively compact in the complete target. Hence RQ is compact.

Conversely if M is infinite dimensional, choose an orthonormal sequence $u_n \in M$. The scalar inputs $X(u_n)$ have norm one, are observed, and $RQX(u_n) = u_n$, viewed as constant fields. The outputs are pairwise at distance $\sqrt{2}$, contradicting compactness. This proves necessity and completes both classifications. □

Source correspondence: `GaussianConditionalCompactness.conditionalProjection_compact_iff`; `GaussianFiniteObservationRecovery.boundedHermiteSpace_dimension`, `observed_chaos_block_compact`, `recovery_compact_iff_finite_source`; `GaussianRecoverySpectralCompactness.recovery_tail_operator_bound`.

Even compact one-source recovery has infinite orthonormal image energy. For unit $u \in M$, its normalized Wick inputs satisfy

$$\left\| RQ \frac{W_q(u)}{\sqrt{q!}} \right\|^2 = \frac{1}{q} \quad (q \ge 1).$$

Their squared image norms have divergent harmonic sum. In particular RQ is not Hilbert–Schmidt whenever $M \neq 0$, including $m = 1$. Decay of each image does not give summability of all images. This is the explicit formal harmonic-energy obstruction, not a new compactness principle.

Source correspondence: `GaussianObservedRecoveryHarmonicEnergy.observed_normalized_energy`, `observed_energy_not_summable`, `one_source_compact_divergent_energy`.

For infinite-dimensional M there is a stronger exact quantitative boundary:

$$\inf\{\|RQ - K\| : K : \mathcal{G} \to \mathcal{V} \text{ compact}\} = 1. \tag{28}$$

Indeed $X(u_n)$ above is orthonormal and hence weakly tends to zero by Bessel's inequality. A compact operator sends this sequence to zero in norm: otherwise a nonvanishing subsequence would have a norm convergent image subsequence, whose limit must be zero by weak continuity, a contradiction. Therefore $\|(RQ - K)X(u_n)\| \ge 1 - \|KX(u_n)\| \to 1$ and $\|RQ - K\| \ge 1$. Taking $K = 0$ and $\|RQ\| \le 1$ proves (28). The frozen source states this distance for the full recovery R; its observed infinite-dimensional version here is an explicitly labelled human deduction from the same first-chaos isometry argument.

Source correspondence: `GaussianRecoveryCompactApproximationGap.compact_approximation_gap`, `optimal_compact_error` (full recovery).

12 An exact positive-chaos criterion for general inputs

The preceding classification concerns a Gaussian conditional projection. There is also a general operator statement that separates chaos-tail decay from compactness inside each fixed chaos.

Theorem 12.1. *Let E be a real normed space and $K : E \to \mathcal{G}$ a bounded linear operator. Then*

$$RK \text{ is compact} \quad \Longleftrightarrow \quad P_q K \text{ is compact for every integer } q \geq 1.$$

No compactness condition on the constant component is required.

Proof. Polarizing (22) gives $R^*R = \sum_{q\geq 1} q^{-1} P_q = N^{-1}$ as a bounded operator. For fixed $q \geq 1$, define the bounded extraction $A_q = qP_qR^* : \mathcal{V} \to \mathcal{G}$. Then $A_qR = P_q$. Thus compactness of RK implies compactness of $P_qK = A_qRK$, since bounded maps preserve relative compactness.

Conversely suppose every P_qK, $q \geq 1$, is compact. The finite sum $RS_NK = \sum_{q=1}^{N} RP_qK$ is compact. The same tail proof as (27), now applied to Kx, gives

$$\|RK - RS_NK\| \leq (N+1)^{-1/2}\|K\| \longrightarrow 0.$$

Norm closure of the compact operators proves the result. Constants are annihilated by R, explaining why they do not appear in the criterion. No finite-dimensionality of a full chaos is assumed here. □

Source correspondence: `GaussianRecoverySpectralCompactness.sectorExtract_recovery`, `partial_composition_compact`, `recovery_composition_compact_iff`.

The theorem explains why inverse-number damping can have a small high-chaos tail while retaining an infinite first-chaos obstruction. It also explains finite-source compactness: each observed fixed-chaos block has a finite Hermite range. These are classical Gaussian spectral mechanisms. Baudoin's primary lecture [2] reconstructs the chaos decomposition and derivative-energy identity underlying (22); its Hermite convention includes $1/q!$ in the polynomial, whereas the present W_q is monic. The exact operator classifications and their formal source reconstruction are presented without a priority claim. Neither compactness nor its failure supplies new fractional-time regularity, curvature or an observed-process stochastic representation.

13 Integrated history: a sufficient state need not compress recovery

The compactness theorem concerns all observed inputs. A finite state sufficient for one terminal target does not reduce that whole input space to finitely many source directions. Integrated Brownian motion gives an exact positive example, with a simultaneous failure of prediction from its scalar current value.

Use the actual time-source space $L^2(\mathbb{R}, dx)$, isometrically embedded in the Gaussian coefficient space as in the preceding constructions. For $t \geq 0$ put

$$b_t = 1_{(0,t]}, \qquad k_t(x) = (t-x)1_{(0,t]}(x), \qquad B_t = X(b_t), \quad J_t = X(k_t).$$

This is the usual integrated Brownian Gaussian family. The map $r \mapsto b_r$ is continuous in L^2 and has norm $\sqrt{r}$, so it is Bochner integrable on each bounded time interval. Indeed $\int_0^t b_r\, dr = k_t$ as a Bochner L^2 integral, by integration on the finite rectangle $0 \leq x, r \leq t$: at fixed x its length is $t - x$ for $0 < x \leq t$, and zero otherwise. The isometry X commutes with this integral, giving

$J_t = \int_0^t B_r\,dr$ in L^2. This elementary identification is a human reconstruction of the formal kernel realization. For $0 \le r \le t$, direct integration gives

$$\mathbb{E}[J_t J_r] = \int_0^r (t-x)(r-x)\,dx = \frac{tr^2}{2} - \frac{r^3}{6}, \qquad \mathbb{E}[J_r^2] = \frac{r^3}{3}. \tag{29}$$

First, the integrated history recovers Brownian state in the closed Gaussian source span. For $0 \le a < s$, let $h = s - a$. The quotient $(k_s - k_a)/h$ equals b_s on $(0, a]$, while on $(a, s]$ its difference from b_s is $-(x-a)/h$. Consequently

$$\left\| \frac{k_s - k_a}{s-a} - b_s \right\|^2 = \frac{1}{h^2}\int_a^s (x-a)^2\,dx = \frac{h}{3}. \tag{30}$$

The source explicitly proves the sufficient bound by h; the exact $h/3$ evaluation here is a labelled classical human refinement. Taking $a = sn/(n+1)$ gives convergence to b_s. The same argument at every $r \le s$ places b_r in the closed span of $\{k_v : 0 \le v \le s\}$. At $r = 0$ the vector is zero. Thus the Brownian state is an L^2-measurable function of the integrated history up to that time, with no terminal observation added. This is a statement about completed L^2 observation spaces; literal raw sigma-algebra equality is not asserted.

Let Q_s^J condition on $\sigma(J_r : 0 \le r \le s)$, and $Q_s^{(2)}$ on $\sigma(J_s, B_s)$. For $s \le t$, put $h = t - s$. The exact source decomposition is

$$k_t = k_s + hb_s + (t-x)1_{(s,t]}(x).$$

The last term is orthogonal to every source supported in $(0, s]$. The forecast vector $k_s + hb_s$ lies both in the integrated-history closed span, by (30), and in the two-state span. It follows that

$$Q_s^J J_t = Q_s^{(2)} J_t = J_s + hB_s, \qquad Q_s^J W_q(k_t) = Q_s^{(2)} W_q(k_t) = W_q(k_s + hb_s). \tag{31}$$

Hermite completeness for the one-dimensional Gaussian J_t, and contraction of both projections, extend the identity to

$$Q_s^J f = Q_s^{(2)} f \qquad (f \in L^2(\sigma(J_t), P)). \tag{32}$$

For $t = 0$ only constants occur; otherwise rescale k_t to unit norm and apply the expansion of Section 1. Hence the entire terminal L^2 space has a two-coordinate sufficient state, not just the linear terminal value.

The scalar current value J_s does not suffice. At $s = 2, t = 3$, (29) gives its scalar regression $\mathbb{E}[J_3 \mid J_2] = \frac{7}{4}J_2$. Testing against the earlier observation J_1 gives

$$\mathbb{E}[(Q_2^J J_3)J_1] = \mathbb{E}[J_3 J_1] = \frac{4}{3}, \qquad \mathbb{E}\left[\left(\frac{7}{4}J_2\right)J_1\right] = \frac{35}{24}.$$

The first equality uses orthogonality of the conditional residual to the history. Since the two numbers differ, the conditional forecasts differ. A further classical calculation quantifies the loss: for $s > 0, h = t - s \ge 0$, the two-state prediction error has variance $h^3/3$, the squared norm of the future kernel in the decomposition. The scalar regression coefficient is $1 + 3h/(2s)$, so its error variance, by subtracting squared covariance divided by $s^3/3$, is $sh^2/4 + h^3/3$. The extra term is strictly positive for $h > 0$.

Nevertheless the whole integrated-history source span is infinite dimensional whenever $s > 0$. It contains every b_r, $r \le s$, and therefore contains the normalized indicators of the disjoint intervals $(s2^{-j-1}, s2^{-j}]$, $j \ge 0$. They are orthonormal. Their first-chaos Gaussian observables are fixed by

Q_s^J, and recovery maps them to the corresponding deterministic unit source fields. These images have pairwise distance $\sqrt{2}$, so RQ_s^J is not compact. In contrast the Gram determinant of k_s, b_s is

$$\|k_s\|^2\|b_s\|^2 - \langle k_s, b_s\rangle^2 = \frac{s^4}{12} > 0.$$

The current-state source span has dimension two; Theorem 11.1 therefore makes $RQ_s^{(2)}$ compact. These recovery conclusions are explicit human applications of the already proved source-dimension criterion. They are compatible with (32), which agrees only on the specified terminal subspace. Neither all observed inputs nor the full recovery operators have been identified.

Integrated Gaussian prediction is classical. Lachal's primary author record [7] describes iterated Brownian integrals and their prediction problems; only that record's stated context, not an uninspected bridge theorem, is used here. No new state-augmentation or Markov principle is claimed. This integrated Brownian family is also distinct from iterated Brownian motion with a random clock.

Source correspondence: `IntegratedBrownianStatePrediction` and `IntegratedBrownianHistoryReconstruction` give actual source projections, all-terminal-L^2 state sufficiency and the scalar counterexample. `BrownianHistoryRecoveryCompactness` constructs the disjoint Brownian source directions; its adaptation to integrated history uses the displayed recovered-state inclusion.

14 A continuous observation with instantaneous information arrival

The geometry of an observation space must be checked independently of continuity of the observed paths. The following classical example also explains why a raw natural filtration cannot silently be replaced by its right-continuous enlargement. It concerns a separate Gaussian observation, not an assertion about the IBM natural filtration.

Let $G = X(u)$ for a unit source $u \in \mathcal{H}$, so G is standard normal. For real t put

$$Y_t = \max(t, 0)G, \qquad \mathcal{F}_t = \sigma(Y_s : s \le t), \qquad M_t = \mathbf{1}_{\{t>0\}}G.$$

Every path of Y is continuous. Nevertheless

$$\mathcal{F}_t = \begin{cases} \{\varnothing, \Omega\}, & t \le 0, \\ \sigma(G), & t > 0. \end{cases}$$

Indeed all observations before zero vanish; for positive t, all past observations are functions of G and $G = Y_t/t$. In particular $\mathcal{F}_{0+} = \bigcap_{t>0} \mathcal{F}_t = \sigma(G)$ differs from $\mathcal{F}_0$, since $\mathbb{P}(G > 0) = 1/2$.

Proposition 14.1. *The process M is a square-integrable martingale for the raw filtration $\mathcal{F}$, with mean zero at every time. For every $t > 0$,*

$$\mathbb{E}|M_t - M_0|^2 = 1.$$

It has no modification whose paths are almost surely right-continuous at zero. If Π_t denotes conditional expectation onto $\mathcal{F}_t$ as an operator on ambient L^2, then

$$\|\Pi_t - \Pi_0\|_{L^2 \to L^2} = 1 \quad (t > 0), \qquad \|Y_t\|_2 = t \longrightarrow 0 \quad (t \downarrow 0).$$

Proof. Adaptedness and square integrability follow from the displayed filtration and $\mathbb{E}G^2 = 1$. For $s \le t$, if $s > 0$ then $M_s = M_t = G$ and this variable is $\mathcal{F}_s$-measurable. If $s \le 0$, conditional expectation onto the trivial sigma algebra is the scalar mean, which is zero whether $M_t = 0$ or $M_t = G$. These two cases prove the martingale property. The increment and observation norm identities follow directly.

Let $\widetilde{M}$ be any modification, so $\widetilde{M}_t = M_t$ almost surely for each fixed t. Intersect these full-measure events only at $t = 0$ and $t = 1/n$, $n \ge 1$. On their common event, $\widetilde{M}_0 = 0$ and $\widetilde{M}_{1/n} = G$ for every n. Right continuity at zero would force $G = 0$. The latter has probability zero under the actual standard Gaussian law. Thus no right-continuous modification exists; no uncountable intersection of exceptional sets has been used.

For positive t, Π_t is the orthogonal projection onto $L^2(\sigma(G))$, while Π_0 projects onto constants. These closed spaces are nested; hence $\Pi_t - \Pi_0$ is the orthogonal projection onto the mean-zero part of $L^2(\sigma(G))$ and has norm at most one. Its value on the unit vector G is G, proving the exact norm one. □

Completing the raw sigma algebras by null sets does not change these conditional expectation operators: the time-zero space still contains only almost surely constant variables. By contrast, right-continuous enlargement inserts G into the time-zero information. In that enlarged filtration the constant process $M_t^+ = G$ for $t \ge 0$ is a continuous martingale, but is not a modification of M because its time-zero value differs almost surely. Thus a regularization that changes the initial information has changed the prediction problem.

The raw-filtration and martingale assertions are mapped to

`ContinuousObservationFiltrationObstruction`.

The exact projection norm and the completion/enlargement explanation are labelled classical human deductions proved above. The mechanism itself is elementary and established: Pitera's author lecture notes [6], Section 2.3, footnote 7 on printed page 10, use $Y_t = tZ$ with a uniform nondegenerate Z to demonstrate the same filtration issue. Gaussianity supplies our particular L^2 normalization, not a new filtration principle. Neither a continuous sample path nor a vanishing observation amplitude proves continuous arrival of information.

The increment energy need not be a time measure

For this same actual raw martingale, the origin increment energy is $\mathbb{E}|M_t - M_0|^2 = 1$ for every $t > 0$. There is no countably additive measure ρ on $\mathbb{R}$ such that

$$\rho((0,t]) = \mathbb{E}|M_t - M_0|^2 \quad (t > 0).$$

Indeed $E_n = (0, 1/(n+1)]$ decreases to the empty set, while $\rho(E_n) = 1$. The first interval has finite mass, so continuity from above gives $\rho(E_n) \to \rho(\varnothing) = 0$, a contradiction. No finiteness of the whole measure is required. Placing an atom at zero does not repair these values: zero belongs to none of E_n. This conclusion concerns the specified usual half-open interval convention and the original time-zero value.

Consequently an energy measure used to complete physical-time step integrands is a genuine additional obligation in a raw filtration. Square integrability and the martingale property alone do not supply it in this generality. This does not contradict constructions under right-continuity or other hypotheses proving the required measure. The continuity-from-above proof is elementary; the actual source correspondences are

`RawMartingaleEnergyMeasureObstruction`.

15 An accelerated Gaussian kernel limit with an initial jump

The preceding observation example retains information instantly. A separate, equally classical limit example loses all state information instantly. It concerns genuine Gaussian Markov kernels rather than conditional projection onto a nested observation family. Let $\gamma = N(0,1)$ and, for $u \geq 0$, define

$$K_u(x,\cdot) = \mathrm{Law}\big(e^{-u}x + \sqrt{1-e^{-2u}}Z\big), \qquad Z \sim \gamma.$$

The displayed affine map is jointly Borel in (x,Z), so it defines a measurable probability kernel. Its action preserves $C_b(\mathbb{R})$ and contracts the supremum norm. For the first assertion, if $x_j \to x$ then the affine argument converges pointwise in Z; bounded convergence applies to each $f \in C_b(\mathbb{R})$. The contraction bound follows by integrating $|f| \leq \|f\|_\infty$.

Two successive transitions, with independent normals Z_1, Z_2, give

$$e^{-(u+v)}x + e^{-v}\sqrt{1-e^{-2u}}Z_1 + \sqrt{1-e^{-2v}}Z_2.$$

The last two terms form a centered normal of variance $e^{-2v}(1-e^{-2u})+1-e^{-2v} = 1-e^{-2(u+v)}$. Thus $K_{u+v} = K_u K_v$, with $K_0(x,\cdot) = \delta_x$. Starting from an independent standard normal x instead gives variance $e^{-2u} + 1 - e^{-2u} = 1$, proving invariance of γ. Fubini is justified for bounded observables throughout. These are ordinary Mehler kernels; no new kernel construction is claimed.

Set $r_n = n+1$ and $K_t^{(n)} = K_{r_n t}$ for $t \geq 0$. At fixed n, x the row is weakly continuous in t: the affine argument is continuous and bounded continuous tests admit dominated convergence. For every fixed x, t, on the other hand,

$$K_t^{(n)}(x,\cdot) \implies J_t(x,\cdot), \qquad J_t(x,\cdot) = \begin{cases} \delta_x, & t = 0, \\ \gamma, & t > 0. \end{cases} \tag{33}$$

At positive t the two affine coefficients converge to zero and one, respectively. A common Z then proves the weak convergence by bounded convergence. At zero all rows are exactly δ_x. The limit still obeys the kernel semigroup law: if either time is zero use the identity kernel; if both are positive both compositions are the constant kernel γ. It also preserves γ. These algebraic properties have not supplied continuity in time.

Proposition 15.1. *On the Banach space $C_b(\mathbb{R})$ with its supremum norm, put*

$$S_t f(x) = \int f(y) J_t(x, dy).$$

The norm-continuity domain at zero and the right norm-derivative domain are both precisely the constant functions. On that domain the derivative is zero. In particular this right-derivative domain is closed and is not dense. For the bounded test $f(x) = \cos x$,

$$\|S_t f - f\|_\infty = 1 + e^{-1/2} \qquad (t > 0).$$

Proof. Let $Pf = (\int f\, d\gamma)\mathbf{1}$. Then P is a bounded idempotent and $S_0 = I$, $S_t = P$ for positive t. For $t > 0$ the norm difference from f is the fixed number $\|Pf - f\|_\infty$. It tends to zero exactly when $Pf = f$, which is equivalent to f being constant. Similarly the difference quotient is $(Pf - f)/t$. A finite norm limit forces $Pf = f$ and then that quotient vanishes. The constants are closed under uniform limits and form a proper subspace, so are not dense. Finally the Gaussian characteristic function gives $P\cos = e^{-1/2}$. The bound $|e^{-1/2} - \cos x| \leq 1 + e^{-1/2}$ is attained at $x = \pi$, proving the exact norm. □

Even pointwise temporal continuity fails at $x = 0$, since $S_0 \cos(0) = 1$ and $S_t \cos(0) = e^{-1/2}$ for positive t. The approximating-row continuity above is weak row continuity, not strong continuity on the full supremum-norm space C_b. Classical Mehler semigroups themselves require this topological care; see Lunardi–Pallara [1], Section 3. The right norm derivative just computed is not a mixed-topology or bounded-pointwise generator.

A full-past Markov realization without continuous paths

There nevertheless is an actual coordinate probability realization. On $\Omega = \mathbb{R}^{[0,\infty)}$ with its product sigma algebra take

$$P_x = \delta_x \text{ at coordinate } 0 \quad \otimes \quad \bigotimes_{t>0} \gamma, \qquad Z_t(\omega) = \omega(t).$$

Here the arbitrary product probability theorem is the standard existence input. Evaluation coordinates are measurable. Finite restrictions have the prescribed independent product laws, so $Z_0 = x$ almost surely and $Z_t \sim \gamma$ for $t > 0$. For $0 \le s < t$, Z_t is independent of the entire raw past $\mathcal{H}_s = \sigma(Z_r : 0 \le r \le s)$, not merely of each individual past coordinate. To see this, independence first holds against finite-coordinate cylinder events by the product law. Those cylinders are a pi-system generating $\mathcal{H}_s$; the class of past events satisfying the independence identity is a lambda-system. The pi-lambda theorem proves the claim. Consequently, for every bounded Borel f,

$$\mathbb{E}_x[f(Z_t) \mid \mathcal{H}_s] = \int f \, d\gamma = \int f(y) J_{t-s}(Z_s, dy) \qquad (s < t).$$

For $s = t$ this follows from adaptedness and $J_0 = I$. Thus the full-past Markov identity holds despite the temporal jump. There is also an explicit conditional distribution given the whole past path. If $p \in \mathbb{R}^{[0,s]}$, set $Q_{s,t}(p, \cdot) = \gamma$ for $t > s$ and $Q_{s,s}(p, \cdot) = \delta_{p(s)}$. This is a measurable kernel, since evaluation at s is measurable. For $t > s$ the joint law of the past path and Z_t is the product of the past law and γ, by the same cylinder argument. For $t = s$ the future coordinate is its past evaluation. These identities prove the conditional-distribution claim directly, without a general disintegration theorem on an uncountable product. The source uses real-indexed coordinates; restricting its past to $[0, s]$ gives the version here by the same independence proof.

No realization with these one-time laws can have almost surely right-continuous paths at zero. Indeed take $t_j = 1/j$. For a fixed start x, choose $\epsilon > 0$ such that $\gamma(\{y : |y - x| > \epsilon\}) > 0$; any finite $\epsilon > 0$ works for a nondegenerate Gaussian. Then in every realization

$$P_x(|Z_{t_j} - x| > \epsilon) = \gamma(\{y : |y - x| > \epsilon\}) > 0$$

for every j. Thus convergence to x in probability fails. Almost sure right continuity with $Z_0 = x$ would imply almost sure, hence in-probability, convergence on this countable sequence, a contradiction. A coordinatewise modification preserves these laws, so cannot repair the obstruction. The product construction asserts coordinate measurability and the stated full-past laws; joint measurability in time and sample is not assumed. A valid kernel semigroup, a Markov conditional law and weak row limits are therefore insufficient by themselves for a continuous-path limit or a dense strong-generator domain. These are classical diagnostic mechanisms, not a new general semigroup convergence theorem.

16 Formal scope and primary comparisons

The verified baseline is `bab688bb6140cd6316c80680095a914ea0d346db`. The core correspondence is `GaussianWickConditionalProjection`, `GaussianTerminalPredictionSpectrum` and

GaussianPredictionRankAsymptotic. Actual continuous-past identification is in ContinuousFractionalPrediction; the two-regime obstruction is FractionalDualRegimePrediction. The inverse-number application is GaussianPotentialPredictionRank, with the actual operator in GaussianInverseNumberPotential. Raw quartic conditional energies and cross-target residuals are in GaussianQuarticConditionalVariance and GaussianQuarticResidualCovariance. The quarter-Hurst witness is FractionalCurrentObservationObstruction. The terminal bracket application is mapped to GaussianTerminalBracketPrediction, GaussianTerminalWeightedBracket, GaussianTerminalBracketObstruction, GaussianTerminalBracketProducts and GaussianTerminalStrongBracket. Its harmonic refinements are separately labelled human deductions. Exact declarations, original hypotheses, source hashes and human proof reorganizations accompany the manuscript. Sections 9–12 add the original joint source/sample recovery identity, the exact finite-source compactness classification and the positive-chaos block criterion. The combined finite-rank estimate and observed distance-one conclusion are labelled human deductions. Fifty exact finite checks now pass in retained and new groups. A further 42 exact integrated-state covariance, regression and quotient checks pass; their infinite-domain conclusions use the printed proofs. verify_recovery.py checks conditional Wick defects through degree eight and two-coordinate derivative/recovery energies through degree four. Infinite compactness and density arguments rest on the printed proofs. The elementary noncompactness of the full potential is a classical deduction from its exact first-chaos identity, not a newly named formal theorem. McElroy–Das's primary introduction describes the Hermite predictor and mean-square-error mechanism for finite data; access to its complete main-theorem text was incomplete, so no detailed theorem-number priority comparison is claimed. Sottinen–Viitasaari's accessible primary paper, Theorem 3.1 and Proposition 3.2, gives the conditional law and the two small-past covariance asymptotics. Our proof retains a weaker explicit uniform lower constant and does not assert a new fractional exponent. The methods and central structure are classical. This is a consolidated formalization companion with explicit domain and regularity diagnostics; novelty remains unresolved. It neither completes the paused Bakry/fBM campaign nor proves new curvature, temporal stochastic integration or neural-network performance claims. The Mehler implementation retains the Michael R. Douglas adaptation credit and Apache-2.0 notice in its original source; see LICENSES/Apache-2.0.txt and MATHEMATICAL_AUDIT.md. Full baseline verification evidence was hash-checked without repeating a research build.

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